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Approximate Nonlinear Filters

1. INTRODUCTION

In nonlinear filtering problems as well as in the linear ones, we are interested in computing the conditional mean and covariance matrix. The conditional mean is our minimum variance estimate, and the covariance matrix measures the uncertainty in the estimate. Throughout our development of nonlinear filter theory in Chapter 6 (see, for example, the discussion following Theorem 6.2), we observed that, in general, the equations of evolution for the conditional mean and covariance matrix depend on all the moments of the conditional density function. In general, the conditional density cannot be characterized by a finite set of parameters (e.g., moments). A notable exception to this, of course, is the linear filtering problem. In that situation, the conditional density is Gaussian and is, therefore, completely determined by its mean vector and covariance matrix. The mean and covariance matrix may be thought of as the *state* of the linear filter. In the nonlinear situation, the state of the filter is essentially infinite, consisting of the whole conditional density function. If we had the conditional density in hand, we could, of course, compute any moment.

We saw in Chapter 6 that numerical problems in computing the whole conditional density function are, in general, intractable, since they involve solution of partial integro-differential equations (Theo-

rem 6.5) or functional integral difference equations (Theorem 6.1). Thus, we are forced to consider approximations to the conditional density function. We should like to *parametrize* the conditional density via a finite and small set of parameters. Our nonlinear filter would then consist of equations of evolution for these parameters, which would comprise the *state* of our filter. If we could find a set of parameters that completely determine the conditional density, these parameters would be *sufficient statistics*. Unfortunately, it is virtually impossible to find sufficient statistics for nonlinear problems. The linear case is a unique exception.

This chapter is devoted to a study of several approximations for the conditional density function. A number of approximate nonlinear filters are developed. The reader will recall that one such approximation was already used in Section 3.9 and Example 4.21 in obtaining approximate equations of evolution for unconditioned moments. We critically discuss these approximations, noting the effects of nonlinearities in nonlinear systems, and describe results of numerical simulations. The approximate nonlinear filters are also compared with the extended Kalman filter and the local iteration algorithms developed in Chapter 8. Finally, we present simulation results in a simple, but nonlinear, problem for which we can actually compute the conditional density function, and describe certain applications in orbit determination and reentry. The chapter closes with a summary that attempts to evaluate nonlinear filtering work, its value and prospects.

2. APPROXIMATION TECHNIQUES: PARAMETRIZATION OF DENSITY FUNCTIONS

A parametrization of the density function which immediately suggests itself is an *orthogonal series* expansion. Restricting ourselves, for simplicity, to scalar systems, we might approximate the conditional density by a (finite) series expansion

$$p(x_t \mid Y_t) \approx \sum_{i=1}^{N} c_i \varphi_i(x), \tag{9.1}$$

where

$$\int \varphi_i(x)\, \varphi_j(x)\, dx = \delta_{ij},$$

so that the coefficients c_i are given by

$$c_i = \int \varphi_i(x)\, p(x \mid Y_t)\, dx.$$

Equations of evolution for the coefficients c_i can be determined from the law of evolution of the conditional density function. The filter then consists of the equations for these coefficients. In view of (9.1), the conditional density can always be reconstructed, and its moments can be computed.

The problem, of course, is to find an appropriate approximating series. An apparently useful series is the *Edgeworth* series (see Cramer [8]). This series, consisting of Hermite polynomials, is useful for nearly Gaussian densities. Developed originally for scalar random variables, it has been extended to the vector case by Kuznetsov *et al.* [20]. The coefficients c_i in this series are called *quasi-moments* by Stratonovich. The Edgeworth series expansion has been used by Sorenson and Stubberud [25] in developing approximate filters for the scalar, discrete system. They had difficulties in developing difference equations for the quasi-moments across an observation (through Bayes' rule) and had to rely on an additional expansion to which the filter was very sensitive. A method of propagating moments through Bayes' rule is described in Section 4, and discussion of Sorenson and Stubberud's results is deferred to Section 6. Quasi-moments have also been used by Fisher [11] in developing an approximate filter for continuous nonlinear systems. His filter is equivalent to one which we develop in Section 3.

A related method of parametrizing the conditional density is *via moments*. Now clearly it is not feasible to develop many moments, as it is not feasible to develop many terms in the expansion (9.1). Thus, it is necessary to truncate the moments or to make some other approximation. In Section 3.9 and Example 4.21, we used the approximation of neglecting third- and higher-order central moments. This sort of approximation is appropriate if the density is (almost) symmetric and concentrated close to the mean (small variance). Another approximation is to suppose that the density is nearly Gaussian. In that case, all the odd central moments can be neglected, and higher-order even moments can be written in terms of the variance. We use these moment approximations to develop nonlinear filters in Sections 3 and 4, and discussion of contributions in this area is deferred to those sections.

Now quasi-moments are closely related to moments. In fact, the first two quasi-moments are nothing more than the mean and variance, and the third quasi-moment is proportional to the third central moment. The fourth quasi-moment q_4 is given by

$$q_4 = (m_4/\sigma^4) - 3, \tag{9.2}$$

where m_4 is the fourth central moment and σ^2 is the variance. In general, if $q_i = 0$, $i \geqslant 3$, the density is Gaussian. As a consequence, an expansion

using Edgeworth-type series and quasi-moments, which is truncated at q_2, is equivalent to our parametrization via moments if it is assumed that the density is nearly Gaussian. Now in problems of any sizable dimension, it is not feasible to carry third and higher order moments or quasi-moments. Therefore, in practice, there is little to be gained in using quasi-moments. For this reason, we will not consider quasi-moments further, but will deal with moment approximation.

An objection to the approximations we have described is that, by neglecting high-order moments or quasi-moments, the approximate density function which results is not really a density function. It can, for instance, assume negative values. This point was raised by Kushner [19], and he suggested computing the expectations in the equations for the conditional mean and covariance (Theorems 6.2 and 6.6) on the basis of some assumed density. We in fact do this in Section 8 for a simple scalar problem. In order to do this in a computationally practical way, we have to find density functions that are characterized by a finite and small set of parameters. Unfortunately, this is not a simple matter. An approach suggested by Kushner is via *moment sequences*. Given two moments, m_1 and m_2 say, find $m_3, \ldots, m_N$ so that $\{m_1, \ldots, m_N\}$ define a density function. Kushner shows how moment sequences can be found in the scalar case, but there does not appear to be a vector generalization. In the final analysis, Kushner assumes that the density is Gaussian.

Reference to Theorems 6.2 and 6.6 shows that the right-hand sides of the equations for the conditional mean and covariance matrix involve expectations of nonlinear functions [involving the system function $f(x, t)$ and measurement function $h(x, t)$] with respect to the conditional density. Even if a particular form of the conditional density is assumed, these expectations are difficult to evaluate because of the nonlinearity of f and h. In our approximations (Sections 3–5), we expand these functions to second order and neglect all but the quadratic nonlinearity. Swerling [27, 28] suggests an approximation to these functions of the form

$$h_i(x, t) = \sum_j \sum_k c_{ijk} \varphi_j{}^i(x)\, \psi_k{}^i(t). \tag{9.3}$$

Then the integrals over x might be computed in advance. This is a laborious process, but needs to be done only once. Real-time implementation of the filter would only involve computing linear combinations of the precomputed integrals. Sunahara [26] proposes to replace the nonlinear functions f and h by quasi-linear functions via stochastic linearization. Neither of these techniques appears to have been tested yet in nonlinear filtering problems.

3. NONLINEAR FILTER APPROXIMATIONS: CONTINUOUS FILTER

The exact equations of evolution for the conditional mean and covariance matrix are given in Theorem 6.6. We develop here several approximations to these equations. First we assume that third- and higher-order central moments are negligible. This is appropriate if the conditional density is almost symmetric and concentrated near its mean. The resulting approximate filter will be called the *truncated second-order filter*. In the second approximation, we do not neglect fourth central moments, but, assuming the density is Gaussian, we write them as

$$[(x_i - \hat{x}_i)(x_j - \hat{x}_j)(x_k - \hat{x}_k)(x_l - \hat{x}_l)]\hat{} = P_{jk}P_{il} + P_{jl}P_{ik} + P_{kl}P_{ij}, \tag{9.4}$$

where P is the conditional covariance matrix. Odd central moments and higher-order even central moments are neglected. In this approximation, we carry the system nonlinearities to second order only. This approximate filter will be called the *Gaussian second-order filter.*

In order not to confuse the reader with notational details, we first consider the scalar filter ($n = m = r = 1$). In Eq. (6.100) of Theorem 6.6, we are required to compute the conditional expectation of the function f. We first expand f in Taylor series around the conditional mean

$$f(x, t) = f(\hat{x}, t) + f_x(\hat{x}, t)(x - \hat{x}) + \tfrac{1}{2}f_{xx}(\hat{x}, t)(x - \hat{x})^2 + \cdots. \tag{9.5}$$

Then, taking the conditional expectation in (9.5), we have

$$\hat{f}(x, t) = f(\hat{x}, t) + \tfrac{1}{2}Pf_{xx}(\hat{x}, t) \tag{9.6}$$

for both approximations. Similarly, for both approximations,

$$\widehat{xf} - \hat{x}\hat{f} = Pf_x(\hat{x}, t). \tag{9.7}$$

Similar expressions are obtained for $\hat{h}$ and $(\widehat{xh} - \hat{x}\hat{h})$. Now for the *truncated* second-order filter,

$$\begin{gathered}\widehat{g^2Q} = g^2(\hat{x}, t)\,Q + g_x{}^2(\hat{x}, t)\,PQ + g(\hat{x}, t)\,g_{xx}(\hat{x}, t)\,PQ,\\ (\widehat{x^2h} - \widehat{x^2}\hat{h} - 2\hat{x}\widehat{xh} + 2\hat{x}^2\hat{h}) = -\tfrac{1}{2}P^2h_{xx}(\hat{x}, t),\end{gathered} \tag{9.8}$$

whereas for the *Gaussian* second-order filter,

$$\begin{gathered}\widehat{g^2Q} = g^2(\hat{x}, t)\,Q + g_x{}^2(\hat{x}, t)\,PQ + g(\hat{x}, t)\,g_{xx}(\hat{x}, t)\,PQ + \tfrac{3}{4}g_{xx}^2(\hat{x}, t)\,P^2Q,\\ (\widehat{x^2h} - \widehat{x^2}\hat{h} - 2\hat{x}\widehat{xh} + 2\hat{x}^2\hat{h}) = P^2h_{xx}(\hat{x}, t).\end{gathered} \tag{9.9}$$

The differences between (9.8) and (9.9) arise because the fourth partials of g^2 and x^2h contain the second partials g_{xx} and h_{xx}, respectively. These extra terms are retained in the Gaussian second-order filter since fourth moments are not neglected but rather approximated as in (9.4).

Then, in view of Theorem 6.6, the *truncated* second-order filter becomes

$$\begin{aligned} d\hat{x}_t &= [f(\hat{x}_t, t) + \tfrac{1}{2}P_t f_{xx}(\hat{x}_t, t)]\,dt + P_t h_x(\hat{x}_t, t)\,R^{-1}(t) \\ &\quad \times [dz_t - (h(\hat{x}_t, t) + \tfrac{1}{2}P_t h_{xx}(\hat{x}_t, t))\,dt], \\ dP_t &= [2P_t f_x(\hat{x}_t, t) + \widehat{g^2Q} - P_t h_x(\hat{x}_t, t)\,R^{-1}(t)\,h_x(\hat{x}_t, t)\,P_t]\,dt \\ &\quad - \tfrac{1}{2}P_t^2 h_{xx}(\hat{x}_t, t)\,R^{-1}(t)[dz_t - (h(\hat{x}_t, t) + \tfrac{1}{2}P_t h_{xx}(\hat{x}_t, t))\,dt], \end{aligned} \tag{9.10}$$

with $\widehat{g^2Q}$ given in (9.8). The *Gaussian* second-order filter has the same equation for the conditional mean as the truncated second-order filter, but the variance equation becomes

$$\begin{aligned} dP_t &= [2P_t f_x(\hat{x}_t, t) + \widehat{g^2Q} - P_t h_x(\hat{x}_t, t)\,R^{-1}(t)\,h_x(\hat{x}_t, t)\,P_t]\,dt \\ &\quad + P_t^2 h_{xx}(\hat{x}_t, t)\,R^{-1}(t)[dz_t - (h(\hat{x}_t, t) + \tfrac{1}{2}P_t h_{xx}(\hat{x}_t, t))\,dt], \end{aligned} \tag{9.11}$$

with $\widehat{g^2Q}$ given in (9.9).

The truncated and Gaussian second-order filters for vector systems are given in Appendix 9A.

The truncated second-order filter was developed by Jazwinski [15, 16], and independently by Bass *et al.* [2]. Schwartz [22], Jazwinski [16], and Fisher [11] independently developed the Gaussian second-order filter. Schwartz and Stear [23] have pointed out that the equations for these filters do not satisfy the (sufficient) conditions for existence and uniqueness of solutions, as given in Theorem 4.5, for example. They consequently modified the approximating expansions for the functions f, g, and h somewhat, and obtained slightly different filter equations which satisfy the conditions of Theorem 4.5. It can be shown, however, that both their modified approximating expansion and the one we use here (Taylor series) produce the same computational results (see Schwartz [22]).

Let us make a few observations about the truncated and Gaussian second-order filters. We first note that both filters reduce to the Kalman–Bucy filter (Theorem 7.3) when the dynamics and observations are linear. Second, we note that, if the terms containing the second partial derivatives f_{xx} and h_{xx} are dropped, and if the noise coefficient matrix G is a function of time only, then both filters reduce to the extended

Kalman filter (see Example 9.1). In both filters, as well as in the extended Kalman filter, the equations for the conditional mean and covariance matrix are coupled; it is no longer possible to precompute the filter gain, as it is in the linear case.

Example 9.1 (Extended Kalman Filter). The extended Kalman filter for continuous-discrete (or discrete) systems is given in Theorem 8.1. This filter is the result of the application of the linear Kalman–Bucy filter to a linearized nonlinear system, where the nonlinear system is relinearized after each observation (see Section 8.3). This has the obvious extension to continuous systems, and we sketch that extension here. Differentiating (8.30) and (8.31) with respect to t_{k+1} (call it t), and using the properties of the state transition matrix, we get

$$\begin{aligned} [d\hat{x}_t/dt]_d &= f(\hat{x}_t, t), \\ [dP_t/dt]_d &= FP_t + P_tF^{\mathrm{T}} + GQG^{\mathrm{T}}, \end{aligned} \tag{9.12}$$

where the matrix F is defined in (8.16) and in Appendix 9A. The subscript d in (9.12) indicates that these are the rates due to the dynamics. Using the limiting procedure described in Example 4.18 and in Eqs. (8.32) and (8.33), we get the following rates of change due to observations:

$$\begin{aligned} [d\hat{x}_t/dt]_0 &= P_tM^{\mathrm{T}}R^{-1}(y_t - h(\hat{x}_t, t)), \\ [dP_t/dt]_0 &= -P_tM^{\mathrm{T}}R^{-1}MP_t, \end{aligned} \tag{9.13}$$

where the matrix M is defined in (8.23) and in Appendix 9A. Combining (9.12) and (9.13), we get the (continuous) *extended Kalman filter*:

$$\begin{aligned} d\hat{x}_t/dt &= f(\hat{x}_t, t) + P_tM^{\mathrm{T}}R^{-1}(y_t - h(\hat{x}_t, t)), \\ dP_t/dt &= FP_t + P_tF^{\mathrm{T}} + GQG^{\mathrm{T}} - P_tM^{\mathrm{T}}R^{-1}MP_t. \end{aligned} \tag{9.14}$$

▲

A significant feature of both the truncated and Gaussian second-order filters is the presence of the random forcing term (measurement differential dz_t) in the variance equation. This term is, of course, properly present (see Theorem 6.6). It is disquieting, however, that in the one filter that term enters with a minus sign, and in the other filter with a plus sign. Since the equations for the conditional covariance matrix are not exact, it is possible for the random forcing term to produce a negative variance. This possibility is particularly strong in case the noise inputs to the system are small and the filter operates over long times. In that case, we can learn the state very precisely (small P). These considerations

lead one to consider a compromise between the truncated and Gaussian second-order filters, which we shall call the *modified second-order filter*:

$$\begin{aligned} d\hat{x}_t &= [f(\hat{x}_t, t) + \tfrac{1}{2}P_t f_{xx}(\hat{x}_t, t)]\, dt \\ &\quad + P_t h_x(\hat{x}_t, t)\, R^{-1}(t)[dz_t - (h(\hat{x}_t, t) + \tfrac{1}{2}P_t h_{xx}(\hat{x}_t, t))\, dt], \\ dP_t/dt &= 2P_t f_x(\hat{x}_t, t) + \widehat{g^2 Q} - P_t h_x(\hat{x}_t, t)\, R^{-1}(t)\, h_x(\hat{x}_t, t)\, P_t . \end{aligned} \tag{9.15}$$

In the modified second-order filter, the random forcing term in the variance equation is neglected.

We remind the reader that the differential equations describing the approximate nonlinear filters are (Itô) stochastic differential equations. The coefficients of dz_t are in general random and, therefore, care must be taken in simulating these filters on a computer. It is tacitly assumed that the Itô equations describing the dynamical system (6.64), (6.65) model the physical system of interest (see Section 4.8). One way of simulating the filters is first to transform the filter equations to the Stratonovich form (see Sections 4.6 and 4.7) and then to use standard integration techniques. Also see Kushner [19] in this connection.

Parallel to the probabilistic results in nonlinear filtering, a number of authors have looked at the problem from the statistical point of view, as described in Chapter 5. They applied the various techniques described in the examples of Chapter 7 (e.g., Examples 7.11–7.13) to nonlinear systems and obtained approximate nonlinear filters by a linearization of one form or another. For example, Cox [5–7] and others have derived the extended Kalman filter in this fashion. Detchmendy and Sridhar [10] derived an approximate nonlinear filter using the invariant imbedding technique described in Example 7.13, plus a linearization. If the arbitrary weighting functions used by Detchmendy and Sridhar are given a probabilistic interpretation, then their filter is of the form

$$\begin{aligned} d\hat{x}_t/dt &= f(\hat{x}_t, t) + P_t h_x(\hat{x}_t, t)\, R^{-1}(t)(y_t - h(\hat{x}_t, t)), \\ dP_t/dt &= 2P_t f_x(\hat{x}_t, t) + g^2(\hat{x}_t, t)\, Q - P_t h_x(\hat{x}_t, t)\, R^{-1}(t)\, h_x(\hat{x}_t, t)\, P_t \\ &\quad + P_t^2 h_{xx}(\hat{x}_t, t)\, R^{-1}(t)(y_t - h(\hat{x}_t, t)). \end{aligned} \tag{9.16}$$

There has been, thus far, little simulation experience with the approximate continuous nonlinear filters we have described (with the exception of the extended Kalman filter, which has been used extensively). Schwartz and Stear [24] report simulations of all these filters (truncated second-order, Gaussian second-order, extended Kalman, modified second-order, Detchmendy and Sridhar's statistical filter, and a few others) in scalar systems, but their results are inconclusive. In essence, according to their simulations, the performance of all these filters is

similar, so that no filter can be recommended on the basis of superior performance. The extended Kalman filter, which involves no second partial derivatives, is therefore recommended on the basis of simplicity and smallest computational requirements. In their simulations, Schwartz and Stear unfortunately use very large measurement noise (R), so that measurement nonlinearities are masked by the noise. They use rather small initial errors (P_0), so that the effect of system nonlinearities is reduced (see Section 6). One conclusion that can be drawn from their simulations is that the extended Kalman filter is superior in performance to the linear filter with no relinearization. This fact is, of course, well known (see Section 8.7, for example).

Kushner [19] has simulated a Gaussian-type approximate filter on a second-order system (van der Pol equation) with *linear* measurements. Third moments were neglected, and fourth moments were approximated as in (9.4), but all the system nonlinearities (terms in Taylor series expansion) were carried. (This dynamical system has cubic nonlinearities.) In Kushner's simulations, the measurement noise was large and the initial errors (P_0) were large, so that system nonlinearities were very pronounced. Kushner's filter tracked the system trajectory, whereas the Gaussian second-order filter (in this case equivalent to the truncated second-order and the modified second-order filters) was unstable and diverged. No comparison with other estimators was given. Since linear measurements were used, the variance equation was nonrandom. As a result, possible difficulties due to the presence of a random forcing term in the variance equation could not be assessed.

4. NONLINEAR FILTER APPROXIMATIONS: CONTINUOUS-DISCRETE FILTER

Using the same approximations as were used in Section 3, we develop in this section approximate filters for the continuous-discrete problem. Comparing the exact equations of evolution for the conditional mean and covariance matrix in continuous and continuous-discrete filtering problems (Theorems 6.6 and 6.2, respectively), we see that approximate equations of evolution *between observations* are obtained by simply setting $R^{-1} \equiv 0$ in the approximate continuous filters of Section 3. Restricting ourselves here to the scalar case (the vector case is treated in Appendix 9B), we have from (9.10) that, between observations,

$$\begin{aligned} d\hat{x}_t/dt &= f(\hat{x}_t, t) + \tfrac{1}{2}P_t f_{xx}(\hat{x}_t, t), \\ dP_t/dt &= 2P_t f_x(\hat{x}_t, t) + \widehat{g^2}Q, \end{aligned} \tag{9.17}$$

where $\widehat{g^2Q}$ is given in (9.8) for the *truncated* second-order filter, and in (9.9) for the *Gaussian* second-order filter.

One can obtain approximate difference equations for the conditional mean and covariance matrix *at an observation* by similarly approximating the expectations on the right-hand side of Eq. (6.26) in Theorem 6.2. This is, in fact, the approach taken by Jazwinski [13] and by Sorenson and Stubberud [25]. In order to approximate these expectations, we must approximate the exponential function $p(y_k \mid x_{t_k})$ by some sort of series expansion. It turns out (see Jazwinski [14]) that this approach is not very satisfactory; the filter is very sensitive to this approximation. We therefore pursue another approach due to Kramer [18] and further developed by Jazwinski [16].

From Theorem 6.1, we have the difference equation for the conditional density at an observation

$$p(x_{t_k} \mid Y_{t_k}) = \frac{p(y_k \mid x_{t_k})\, p(x_{t_k} \mid Y_{t_{k-1}})}{p(y_k \mid Y_{t_{k-1}})}, \tag{9.18}$$

which is essentially Bayes' rule. Now

$$p(y_k \mid x_{t_k}) = p(y_k \mid x_{t_k}, Y_{t_{k-1}}),$$

and

$$p(y_k \mid x_{t_k}, Y_{t_{k-1}})\, p(x_{t_k} \mid Y_{t_{k-1}}) = p(x_{t_k}, y_{t_k} \mid Y_{t_{k-1}}).$$

Therefore, (9.18) becomes

$$p(x_{t_k}, y_{t_k} \mid Y_{t_{k-1}}) = p(x_{t_k} \mid Y_{t_k})\, p(y_{t_k} \mid Y_{t_{k-1}}). \tag{9.19}$$

To determine how moments propagate across an observation, we multiply (9.19) by the product of the two functions $s(x_{t_k})\, u(y_{t_k})$ and integrate over x and y:

$$\int\int s(x)\, u(y)\, p(x, y \mid Y_{t_{k-1}})\, dx\, dy = \int\int s(x)\, u(y)\, p(x \mid Y_{t_k})\, p(y \mid Y_{t_{k-1}})\, dx\, dy. \tag{9.20}$$

The right-hand side of (9.20) can be written

$$\int\int s(x)\, u(y)\, p(x \mid Y_{t_k})\, p(y \mid Y_{t_{k-1}})\, dx\, dy$$

$$= \int \left[\int s(x)\, p(x \mid Y_{t_k})\, dx\right] u(y)\, p(y \mid Y_{t_{k-1}})\, dy,$$

and, combining this with (9.20), we have

$$\mathcal{E}\{s(x_{t_k})\, u(y_{t_k}) \mid Y_{t_{k-1}}\} = \mathcal{E}\{\mathcal{E}\{s(x_{t_k}) \mid Y_{t_k}\}\, u(y_{t_k}) \mid Y_{t_{k-1}}\}. \tag{9.21}$$

Equation (9.21) provides the fundamental relationship we are after. The moments we want to compute are of the form

$$\mathcal{E}\{s(x_{t_k}) \mid Y_{t_k}\},$$

so that (9.21) gives us a relationship between (any) moments prior to an observation and after an observation. We use (9.21) in the following way. Suppose we want to compute the conditional mean

$$\hat{x}(t_k \mid t_k) \equiv \mathcal{E}\{x_{t_k} \mid Y_{t_k}\}.$$

We observe that $\hat{x}(t_k \mid t_k)$ is a function of y_{t_k}. If we assume that $\hat{x}(t_k \mid t_k)$ is a power series in y_{t_k}, then (9.21) can be used to determine the coefficients in that power series. In this way, we can obtain approximate difference equations for any moment at an observation. The use of (9.21) is demonstrated in the following example.

Example 9.2 (Kalman Filter Rederived). Suppose observations are linear,

$$y_{t_k} = M(k)\, x_{t_k} + v_{t_k},$$

and assume that the conditional mean and covariance matrix are of the form

$$\begin{aligned} \hat{x}(t_k \mid t_k) &= a + B(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}}), \\ P(t_k \mid t_k) &= C, \end{aligned} \tag{9.22}$$

where

$$\hat{y}_{t_k}^{t_{k-1}} \equiv \mathcal{E}\{y_{t_k} \mid Y_{t_{k-1}}\},$$

and a, B, C are, respectively, $n \times 1$, $n \times m$, $n \times n$ matrices to be determined. We shall rederive the Kalman filter [(7.29) of Theorem 7.2] using (9.21). That is, we shall use (9.21) to determine the coefficients a, B, and C.

Setting $s(x_{t_k}) = x_{t_k}$ and $u(y_{t_k}) = 1$ in (9.21), we have

$$\hat{x}(t_k \mid t_{k-1}) = \mathcal{E}\{\hat{x}(t_k \mid t_k) \mid Y_{t_{k-1}}\}. \tag{9.23}$$

With

$$s = x_{t_k}, \qquad u = (y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})^{\mathrm{T}},$$

(9.21) becomes

$$\mathscr{E}\{x_{t_k}(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})^{\mathrm{T}} \mid Y_{t_{k-1}}\} = \mathscr{E}\{\hat{x}(t_k \mid t_k)(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})^{\mathrm{T}} \mid Y_{t_{k-1}}\}, \tag{9.24}$$

and, with $s = [x_{t_k} - \hat{x}(t_k \mid t_k)][x_{t_k} - \hat{x}(t_k \mid t_k)]^{\mathrm{T}}$, $u = 1$, (9.21) is

$$\mathscr{E}\{[x_{t_k} - \hat{x}(t_k \mid t_k)][x_{t_k} - \hat{x}(t_k \mid t_k)]^{\mathrm{T}} \mid Y_{t_{k-1}}\} = \mathscr{E}\{P(t_k \mid t_k) \mid Y_{t_{k-1}}\}. \tag{9.25}$$

Now, substituting (9.22) in (9.23)–(9.25), we get

$$\hat{x}(t_k \mid t_{k-1}) = a, \tag{9.26}$$

$$\mathscr{E}\{x_{t_k}(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})^{\mathrm{T}} \mid Y_{t_{k-1}}\} = \mathscr{E}\{[a + B(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})][y_{t_k} - \hat{y}_{t_k}^{t_{k-1}}]^{\mathrm{T}} \mid Y_{t_{k-1}}\}, \tag{9.27}$$

$$\mathscr{E}\{[x_{t_k} - a - B(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})][x_{t_k} - a - B(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})]^{\mathrm{T}} \mid Y_{t_{k-1}}\} = C. \tag{9.28}$$

Thus we have already determined the vector a [in (9.26)]. Since

$$y_{t_k} - \hat{y}_{t_k}^{t_{k-1}} = M(k)[x_{t_k} - \hat{x}(t_k \mid t_{k-1})] + v_{t_k},$$

(9.27) becomes

$$P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(k) = B[M(k)\, P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(k) + R(k)]. \tag{9.29}$$

Using (9.26) in (9.28), we get

$$\begin{aligned} &P(t_k \mid t_{k-1}) - P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(k)\, B^{\mathrm{T}} - BM(k)\, P(t_k \mid t_{k-1}) \\ &\quad + B[M(k)\, P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(k) + R(k)]\, B^{\mathrm{T}} = C. \end{aligned} \tag{9.30}$$

Finally, substituting for a, B, and C from (9.26), (9.29), and (9.30), our filter (9.22) becomes

$$\begin{aligned} \hat{x}(t_k \mid t_k) &= \hat{x}(t_k \mid t_{k-1}) + P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(k) \\ &\quad \times [M(k)\, P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(k) + R(k)]^{-1} \\ &\quad \times [y_{t_k} - M(k)\, \hat{x}(t_k \mid t_{k-1})], \\ P(t_k \mid t_k) &= P(t_k \mid t_{k-1}) - P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(k) \\ &\quad \times [M(k) P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(k) + R(k)]^{-1} \\ &\quad \times M(k)\, P(t_k \mid t_{k-1}), \end{aligned} \tag{9.31}$$

which is the Kalman filter. ▲

In developing approximate equations for the conditional mean and covariance at an observation, we assume the power series

$$\begin{aligned} \hat{x}(t_k \mid t_k) &= \sum_{i=0}^{N} a_i (y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})^i, \\ P(t_k \mid t_k) &= \sum_{i=0}^{M} b_i (y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})^i, \end{aligned} \tag{9.32}$$

and use (9.21) to evaluate the coefficients in these series. (We restrict ourselves to the scalar case in this section; the vector case is treated in Appendix 9B.) Using Example 9.2 as a guide, it is easy to see that the following choice of the functions s and u in (9.21) will determine the coefficients:

$$s = x_{t_k}, \qquad u = (y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})^i$$

will determine a_i, $i = 0, 1, \ldots, N$;

$$s = (x_{t_k} - \hat{x}(t_k \mid t_k))^2, \qquad u = (y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})^i$$

will determine b_i, $i = 0, 1, \ldots, M$. If we were interested in the coefficients of the expansion for the third central moment, we would use

$$s = (x_{t_k} - \hat{x}(t_k \mid t_k))^3, \qquad u = (y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})^i,$$

and so on. The extension to the vector case is straightforward, although the notation becomes rather messy.

For practical (computational) reasons, we carry the series in (9.32) to first order only and assume the forms

$$\begin{aligned} \hat{x}(t_k \mid t_k) &= a + B(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}}), \\ P(t_k \mid t_k) &= C + D(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}}). \end{aligned} \tag{9.33}$$

Then from (9.21) we have for the coefficients

$$\begin{aligned} a &= \hat{x}, \\ B &= (\widehat{xh} - \hat{x}\hat{h})[[(h - \hat{h})^2]\hat{} + R]^{-1}, \\ C &= P(t_k \mid t_{k-1}) - (\widehat{xh} - \hat{x}\hat{h})[[(h - \hat{h})^2]\hat{} + R]^{-1}(\widehat{hx} - \hat{h}\hat{x}), \\ D &= \{[(x - \hat{x})^2 h]\hat{} - 2B[(h - \hat{h})(x - \hat{x})h]\hat{} + B^2[[(h - \hat{h})^2 h]\hat{} + R\hat{h}] - C\hat{h}\} \\ &\quad \times \{[(h - \hat{h})^2]\hat{} + R\}^{-1}, \end{aligned} \tag{9.34}$$

where, to save space, we dropped the subscript t_k on x. The measurement function h is evaluated at x_{t_k}, R is $R(k)$, and the caret denotes the expectation using $p(x_{t_k} \mid Y_{t_{k-1}})$, that is,

$$\hat{\varphi} = [\varphi]\hat{} = \mathscr{E}\{\varphi(x_{t_k}) \mid Y_{t_{k-1}}\}.$$

We now approximate the expectations appearing in (9.34) as we did for the continuous filter in Section 3. For both the truncated and Gaussian second-order filters [see (9.7)],

$$\widehat{xh} - \hat{x}\hat{h} = P(t_k \mid t_{k-1})\, h_x(\hat{x}, t_k). \tag{9.35}$$

For the *truncated* second-order filter,

$$[(h - \hat{h})^2]\hat{} = h_x(\hat{x}, t_k)\, P(t_k \mid t_{k-1})\, h_x(\hat{x}, t_k) - \tfrac{1}{4}P^2(t_k \mid t_{k-1})\, h_{xx}^2(\hat{x}, t_k), \tag{9.36}$$

$$\begin{aligned} D = {} & -\tfrac{1}{2}P^2(t_k \mid t_{k-1})\, h_{xx}[[(h - \hat{h})^2]\hat{} + R]^{-1} \\ & \times \{1 - 3P(t_k \mid t_{k-1})\, h_x{}^2[[(h - \hat{h})^2]\hat{} + R]^{-1} \\ & - P(t_k \mid t_{k-1})\, h_x{}^2[[(h - \hat{h})^2]\hat{} + R]^{-2}\,[R + \tfrac{1}{4}P^2(t_k \mid t_{k-1})\, h_{xx}^2] \\ & + 2P^2(t_k \mid t_{k-1})\, h_x{}^4[[(h - \hat{h})^2]\hat{} + R]^{-2}\}, \end{aligned} \tag{9.37}$$

whereas for the *Gaussian* second-order filter,

$$[(h - \hat{h})^2]\hat{} = h_x(\hat{x}, t_k)\, P(t_k \mid t_{k-1})\, h_x(\hat{x}, t_k) + \tfrac{1}{2}P^2(t_k \mid t_{k-1})\, h_{xx}^2(\hat{x}, t_k), \tag{9.38}$$

$$\begin{aligned} D = {} & P^2(t_k \mid t_{k-1})\, h_{xx}[[(h - \hat{h})^2]\hat{} + R]^{-1} \\ & \times \{1 - \tfrac{9}{2}P(t_k \mid t_{k-1})\, h_x{}^2[[(h - \hat{h})^2]\hat{} + R]^{-1} \\ & + \tfrac{1}{2}P(t_k \mid t_{k-1})\, h_x{}^2[[(h - \hat{h})^2]\hat{} + R]^{-2}\,[R + 7P(t_k \mid t_{k-1})\, h_x{}^2]\}. \end{aligned} \tag{9.39}$$

Of course, in both filters,

$$\hat{y}_{t_k}^{t_{k-1}} = h(\hat{x}, t_k) + \tfrac{1}{2}P(t_k \mid t_{k-1})\, h_{xx}(\hat{x}, t_k). \tag{9.40}$$

Let

$$Y^t \triangleq h_x P(t_k \mid t_{k-1})\, h_x + R(k) - \tfrac{1}{4}P^2(t_k \mid t_{k-1})\, h_{xx}^2, \tag{9.41}$$

$$Y^G \triangleq h_x P(t_k \mid t_{k-1})\, h_x + R(k) + \tfrac{1}{2}P^2(t_k \mid t_{k-1})\, h_{xx}^2, \tag{9.42}$$

where h, h_x, and h_{xx} are evaluated at $(\hat{x}(t_k \mid t_{k-1}), t_k)$. Then the *truncated* second-order filter (at an observation) is given by

$$\begin{aligned} \hat{x}(t_k \mid t_k) &= \hat{x}(t_k \mid t_{k-1}) + P(t_k \mid t_{k-1})\, h_x[Y^t]^{-1}\,[y_{t_k} - h - \tfrac{1}{2}P(t_k \mid t_{k-1})\, h_{xx}], \\ P(t_k \mid t_k) &= P(t_k \mid t_{k-1}) - P(t_k \mid t_{k-1})\, h_x[Y^t]^{-1}\, h_x P(t_k \mid t_{k-1}) \\ &\quad + D[y_{t_k} - h - \tfrac{1}{2}P(t_k \mid t_{k-1})\, h_{xx}], \end{aligned} \tag{9.43}$$

where D is given in (9.37). The *Gaussian* second-order filter is given by

$$\begin{aligned}\hat{x}(t_k \mid t_k) &= \hat{x}(t_k \mid t_{k-1}) + P(t_k \mid t_{k-1})\, h_x[Y^G]^{-1}\,[y_{t_k} - h - \tfrac{1}{2}P(t_k \mid t_{k-1})\, h_{xx}],\\ P(t_k \mid t_k) &= P(t_k \mid t_{k-1}) - P(t_k \mid t_{k-1})\, h_x[Y^G]^{-1}\, h_x P(t_k \mid t_{k-1})\\ &\quad + D[y_{t_k} - h - \tfrac{1}{2}P(t_k \mid t_{k-1})\, h_{xx}],\end{aligned} \tag{9.44}$$

with D given in (9.39). We also define the *modified truncated* and *modified Gaussian* second-order filters, which consist of (9.43) and (9.44), respectively, but with $D \equiv 0$. All these filters of course use (9.17) between observations.

We observe that all the approximate nonlinear filters reduce to the Kalman filter (Theorem 7.1) when the dynamics and observations are linear. When the second partial derivatives f_{xx} and h_{xx} are dropped, they reduce to the extended Kalman filter. Using the limiting procedure introduced in Example 4.18, the reader may verify that these approximate continuous-discrete filters limit to their continuous counterparts developed in Section 3.

The remarks made in Section 3 in regard to continuous approximate filters apply here as well, with the following exception. Because observations are discrete, these filters do not involve stochastic differential equations, and there is therefore no problem in simulating them. Some simulations are described in Sections 8–10. Since the Gaussian approximation has a wider range of validity than the truncated approximation (see the first paragraph of Section 3), we do not use the truncated second-order filters in the simulations, but simulate only the Gaussian second-order filters.

We note that the modified Gaussian second-order filter was developed independently by Athans *et al.* [1].

5. NONLINEAR FILTER APPROXIMATIONS: DISCRETE FILTER

We do not develop the details of approximate discrete filters. However, all the ingredients have essentially been derived, and the reader can easily complete the details himself. Equations of evolution for the mean and variance (between observations) of scalar discrete systems are given in Section 3.9. Approximate difference equations at an observation, given in the preceding section, are of course directly applicable to the discrete filter.

Approximations for the nonlinear discrete filtering problem have been

developed by Friedland and Bernstein [12] using statistical methods (Section 5.3). We have already mentioned in Section 4 the work of Sorenson and Stubberud [25]. However, we do not pursue those methods in this book.

6. DISCUSSION OF NONLINEAR EFFECTS

Inspection of the approximate nonlinear filters reveals that the size of nonlinearities depends not only on the size of the second partial derivatives (f_{xx} and h_{xx}), but also on the estimation error variance. The second-order terms appear as

$$Ph_{xx}$$

in the case of measurement nonlinearity, and as

$$Pf_{xx}$$

in the case of system nonlinearity. These are the *expected values* of the second-order terms in the Taylor series expansions. As a consequence, these nonlinear terms can be large because the second partial derivatives are large, or because the estimation error variance is large, or both. We cannot be definitive as to what we mean here by "large" in an absolute sense, but we shall subsequently have something to say about the *significance* of nonlinear effects.

It is important to distinguish between the several ways in which these nonlinear terms can be large. If the nonlinearity is large because the second partial derivatives are large, we call it a *real* nonlinearity. If it is large because the estimation error variance is large, we call it an *induced* nonlinearity. If it is large because the second partials and the error variance are both large, we call it a *mixed* nonlinearity. We note that a nonlinearity can always be induced by setting the initial estimation error variance P_0 sufficiently large.

Now, we expect our approximate filters to be useful and effective when nonlinearities are of the *real* type, that is, when nonlinear effects are significant because the second partials are large while error variances are small. For if the error variance P is small, then P^2, P^4, etc., are smaller still, and the terms that we dropped in our approximations are small relative to the terms we have retained.[1] In case the nonlinearity is induced or mixed, the terms we have neglected can be significant relative to the terms we have retained in these approximations.

[1] To be rigorous, we should be talking here about the magnitude of the remainder in a Taylor series. See footnote 17 in Chapter 3.

Regardless of whether nonlinearities are real, induced, or mixed, it is important to determine when they are *significant.* Denham and Pines [9], Jazwinski [14], Carney and Goldwyn [4], and Sorenson and Stubberud [25] have observed experimentally that measurement nonlinearities become significant when the measurement noise variance R is small while the estimation error variance P is relatively large. Sorenson and Stubberud [25] have also observed that system nonlinearities become significant when the system noise variance Q is small. In general, nonlinear effects appear to be significant when noise inputs are small while the estimation error variance is relatively large. This appears to be rather intuitively obvious, perhaps because of hindsight. Large noise inputs can effectively "cover" neglected nonlinearities.

To assess qualitatively the significance of system nonlinearities, consider Eqs. (9.17), which describe the approximate evolution of the conditional mean and variance between observations. If Q is sufficiently large, then P cannot become very small as a result of information contained in the observations, since we are constantly adding noise to the system. As long as P is large, a small error or bias in the mean $\hat{x}$ is not very significant. [Such a bias in $\hat{x}$ would be produced by neglecting the nonlinear term in the first of Eqs. (9.17).] Now if Q is small and P is initially large but subsequently becomes small, even a small bias in $\hat{x}$ can be significant. The approximate mean can easily fall outside the 3σ limits. Of course if P remains large, significant errors can be made in prediction over long time intervals because of the integrated effect of neglected nonlinearities. Neglected system nonlinearities have a biasing effect on the estimate.

Let us now consider the measurement nonlinearities. We shall look specifically at approximate continuous-discrete filters.

It is difficult to assess the importance of the random forcing term in the variance equation [term with coefficient D in (9.43) or (9.44)], except to observe that its presence can lead to serious difficulties. As was noted in Section 3, that term can produce a negative error variance P. It is easy to see that this random forcing term is significant if $Ph_{xx}/Y^{1/2}$ is comparable to, or larger than, Ph_x^2/Y. Since simulations (Section 8) indeed show that this term can produce negative error variances, we shall restrict our attention to the modified second-order filters.

The forcing term $\frac{1}{2}Ph_{xx}$ in the equation for the conditional mean [in (9.43) or (9.44)] is clearly a (nonlinearity) bias correction term. It ensures that

$$\mathscr{E}\{y_{t_k} - \hat{y}_{t_k}^{t_{k-1}}\} = 0$$

to second order. The significance of this bias correction can be assessed by looking at the variance of

$$(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}}),$$

which is Y^t or Y^G, depending on the approximation used. This will also enable us to assess the significance of nonlinearities in the filter gain itself, since the gain is inversely proportional to Y.

Looking at Eq. (9.41) or (9.42), we see that measurement nonlinearities are significant if

$$P^2 h_{xx}^2 \gtrsim R. \tag{9.45}$$

This observation was also made independently by Denham and Pines [9] and substantiates the experimental findings cited earlier. Measurement nonlinearities are significant if they are comparable to, or larger than, the measurement noise. Significant neglected measurement nonlinearities bias the estimate and result in an incorrect weighting of observations.

Some of the qualitative effects of nonlinearities described in this section will be demonstrated in Section 8 via simulations.

7. LOCAL ITERATIONS

In Section 8.3, we derived two local iteration schemes which, we claim, reduce the (adverse) effect of nonlinearities on the performance of the linear filter applied to a linearized nonlinear system. The iterated extended Kalman filter (Theorem 8.2) reduces the effect of measurement nonlinearities, whereas the iterated linear filter-smoother (Theorem 8.3) reduces the effect of system nonlinearities as well. These iterators were developed as relatively straightforward extensions of the linear theory.

That these iterators are indeed effective in nonlinear problems has been demonstrated experimentally by Denham and Pines [9] and by Wishner *et al.* [29]. We report some simulation experience with these iterators in Sections 8 and 10.

In the present section, we show that these iterators also have a probabilistic interpretation. They are related to maximum likelihood (Bayesian) estimation (see Section 5.3 for definitions of maximum likelihood estimates).

In Example 6.2, we derived the equations for the conditional mode at an observation, assuming, for simplicity of notation, that observations are scalar ($m = 1$). By the conditional mode, we mean the mode of

$$p(x_{t_k} \mid Y_{t_k}).$$

Assuming that $p(x_{t_k} \mid Y_{t_{k-1}})$ is Gaussian with mean $\hat{x}(t_k \mid t_{k-1})$ and covariance matrix $P(t_k \mid t_{k-1})$, the mode of $p(x_{t_k} \mid Y_{t_k})$, call it m_{t_k}, satisfies [from (6.52)]

$$\varphi_x(m_{t_k}) \triangleq -M^{\mathrm{T}}(m_{t_k})\, R^{-1}(k)[y_{t_k} - h(m_{t_k}, t_k)] + P^{-1}(t_k \mid t_{k-1}) \times [m_{t_k} - \hat{x}(t_k \mid t_{k-1})] = 0. \tag{9.46}$$

The matrix of second partial derivatives φ_{xx} is [from (6.53)]

$$\varphi_{xx}(m_{t_k}) = P^{-1}(t_k \mid t_{k-1}) + M^{\mathrm{T}}(m_{t_k})\, R^{-1}(k)\, M(m_{t_k}) - h_{xx}(m_{t_k})\, R^{-1}(k)[y_{t_k} - h(m_{t_k}, t_k)], \tag{9.47}$$

where

$$h_{xx} = \left[\frac{\partial^2 h}{\partial x_i\, \partial x_j}\right] \qquad (n \times n).$$

Suppose we solve (9.46) for the mode using Newton–Raphson iteration.[2] Then our iterator would be

$$\bar{\eta}_{i+1} = \bar{\eta}_i - \varphi_{xx}^{-1}(\bar{\eta}_i)\, \varphi_x(\bar{\eta}_i). \tag{9.48}$$

This iteration involves the computation of the matrix of second partials h_{xx}. Kelley and Denham [17] recommend the Davidon optimization method for computing the mode (maximum likelihood estimate). The Davidon method does not require the second partial derivatives φ_{xx}, yet possesses desirable convergence properties similar to Newton–Raphson iteration.

Let us modify the Newton–Raphson iteration (9.48) by setting $h_{xx} \equiv 0$ in φ_{xx} of (9.47). Then iteration (9.48) becomes

$$\eta_{i+1} = \eta_i + [P^{-1}(t_k \mid t_{k-1}) + M^{\mathrm{T}}(\eta_i)\, R^{-1}(k)\, M(\eta_i)]^{-1} \times [M^{\mathrm{T}}(\eta_i)\, R^{-1}(k)\, (y_{t_k} - h(\eta_i, t_k)) + P^{-1}(t_k \mid t_{k-1})(\hat{x}(t_k \mid t_{k-1}) - \eta_i)]. \tag{9.49}$$

Applying the matrix equalities given in Appendix 7B to the right-hand side of (9.49), we get

$$\eta_{i+1} = \hat{x}(t_k \mid t_{k-1}) + P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(\eta_i) \times [M(\eta_i)\, P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(\eta_i) + R(k)]^{-1} \times [y_{t_k} - h(\eta_i, t_k) + M(\eta_i)(\eta_i - \hat{x}(t_k \mid t_{k-1}))]. \tag{9.50}$$

[2] The reader unfamiliar with this iteration can consult any textbook on numerical analysis.

Comparing this with Theorem 8.2, we see that we have derived the iterator of the iterated extended Kalman filter. Iteration (9.50) [or (9.49)] will clearly converge (locally) to the mode as long as

$$P^{-1}(t_k \mid t_{k-1}) + M^{\mathrm{T}}(\eta_i)\, R^{-1}(k)\, M(\eta_i) > 0.$$

Therefore, the iterated extended Kalman filter consists of the following. Between observations, the conditional mean and covariance matrix propagate according to first-order (not linear) theory [(8.30) and (8.31)]. At an observation, assuming the prior density is Gaussian, this filter solves for the conditional mode of the posterior density. The conditional covariance matrix is then computed according to first-order theory [(8.33)]. The conditional mode is then used for the conditional mean.

Let the reader show that the iterated linear filter-smoother (Theorem 8.3) can be similarly derived by solving for the mode of the conditional density

$$p(x_{t_k}, x_{t_{k-1}} \mid Y_{t_k}).$$

In the two iterated filters just described, the mode of the posterior density is used for the conditional mean. Rauch [21] describes an approximate method of obtaining the mean from the mode by a local expansion in the vicinity of the mode. This expansion is valid when the mean is close to the mode, and the correction is of second order. Breakwell [3] has pointed out that these local iterations produce a *biased* estimate. This is clearly so, since the mode is not equivalent to the mean in nonlinear problems. However, as the error variance becomes small, so does the bias in the estimate. Recall (Section 6) that nonlinearities are proportional to P.

8. A NONLINEAR EXAMPLE

We describe here the results of numerical simulations of some of the approximate nonlinear filters we have developed. Our continuous dynamical system is scalar, but nonlinear, and nonlinear observations are taken at discrete time instants. The system is sufficiently simple so that the conditional probability density function can be computed, thus making it possible to obtain a true evaluation of approximate filter performance. Similar simulations have been previously reported by this author [14].

Our dynamical system consists of the scalar Ricatti equation

$$dx_t/dt = ax_t + bx_t^2 \qquad (a, b \text{ const}), \qquad t \geq t_0, \tag{9.51}$$

and the discrete observations

$$y_{t_k} = x_{t_k}^2 + x_{t_k}^3 + v_k \qquad [h(x) = x^2 + x^3]. \tag{9.52}$$

Since (9.51) is noise-free, we will be able to map the conditional density between observations with the aid of Theorem 2.7. The Ricatti equation (9.51) has the solution

$$\varphi(t, t_0, x_{t_0}) = \frac{a x_{t_0} e^{a(t-t_0)}}{a + b x_{t_0}[1 - e^{a(t-t_0)}]}. \tag{9.53}$$

Consider the function $\varphi(t_{k+1}, t_k, \cdot)$, and let $\psi(t_{k+1}, t_k, \cdot)$ be the inverse function. Clearly,

$$\psi(t_{k+1}, t_k, x) = \frac{a x e^{a(t_k - t_{k+1})}}{a + bx[1 - e^{a(t_k - t_{k+1})}]}. \tag{9.54}$$

Then, according to Theorem 2.7,

$$p(x, t_{k+1}; Y_{t_k}) = p(\psi(t_{k+1}, t_k, x), t_k; Y_{t_k})(\partial\psi/\partial x), \tag{9.55}$$

where we used the notation defined in Sect. 3.2,

$$p(x, t_l; Y_{t_k}) = p(x_{t_l} \mid Y_{t_k}).$$

The difference equation for the conditional density at an observation is of course Eq. (6.14) of Theorem 6.1.

Having described the evolution of the conditional density for our filtering problem, we define the *exact* nonlinear filter. In the exact filter, the conditional mean and variance are computed from the conditional density. That is, the expectations in (6.26) of Theorem 6.2 are computed by numerical quadratures. The result is the *exact* conditional mean and *exact* conditional variance. We note that it is not feasible to compute the exact mean and variance in all but scalar problems, or perhaps two-dimensional problems. We of course have in mind real-time filtering applications when we speak of computational feasibility.

In order to assess the effect of the Gaussian approximation, we introduce what we shall call the *Gaussian* filter. We simulate this filter in static problems only ($\dot{x}_t \equiv 0$). Assuming that the prior density $p(x_{t_k} \mid Y_{t_{k-1}})$ is Gaussian with mean $\hat{x}(t_k \mid t_{k-1})$ and variance $P(t_k \mid t_{k-1})$, this filter computes $\hat{x}(t_k \mid t_k)$ and $P(t_k \mid t_k)$ via Eq. (6.26) of Theorem 6.2.

Simulations of the Gaussian and truncated second-order filters quickly showed that our concern about the presence of a random forcing term in the variance equation was indeed justified. These filters frequently produced negative variances, and we were forced to abandon them.

We present here some simulations of the exact, Gaussian, iterated extended Kalman, modified Gaussian second-order, and extended Kalman filters. Since the exact conditional mean and variance are available (exact filter), we are able to give a true assessment of approximate filter performance. We emphasize that hundreds of simulations were made in the course of this study, with different (random) initial conditions and noise samples. The simulations presented here are representative.

To assess the effect of measurement nonlinearities, we first present simulations for the static problem [$a = b = 0$ in (9.51)]. With $x_{t_0} = 2.0$, $\hat{x}_{t_0} = 2.3$, $P_{t_0} = 0.01$, and observations every 0.1 units of time, starting at 0.1, the various filters are simulated for three values of measurement noise variance R (0.09, 0.01, 0.0001). The results, presented in Figs. 9.1–9.3, enable us to study the significance of measurement nonlinearities.

We first observe that differences between the exact, Gaussian, and iterated extended Kalman filters are negligible, and do not even show on the scale used for plotting in these figures. This means that for our problem the Gaussian approximation is an excellent one. If the conditional density were precisely Gaussian, then the Gaussian and iterated extended Kalman filters would give the exact conditional mean. In

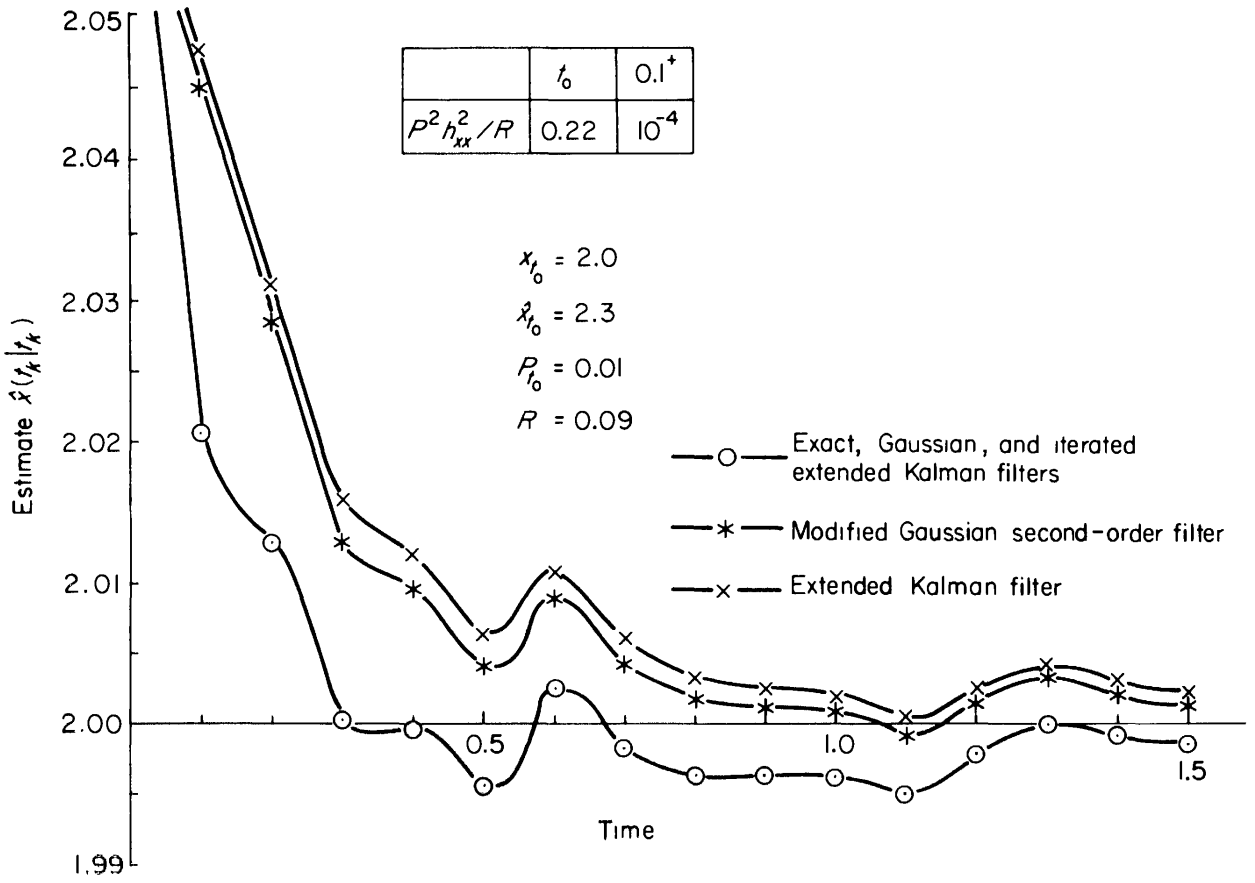


FIG. 9.1. Significance of measurement nonlinearity—I ($R = 0.09$).

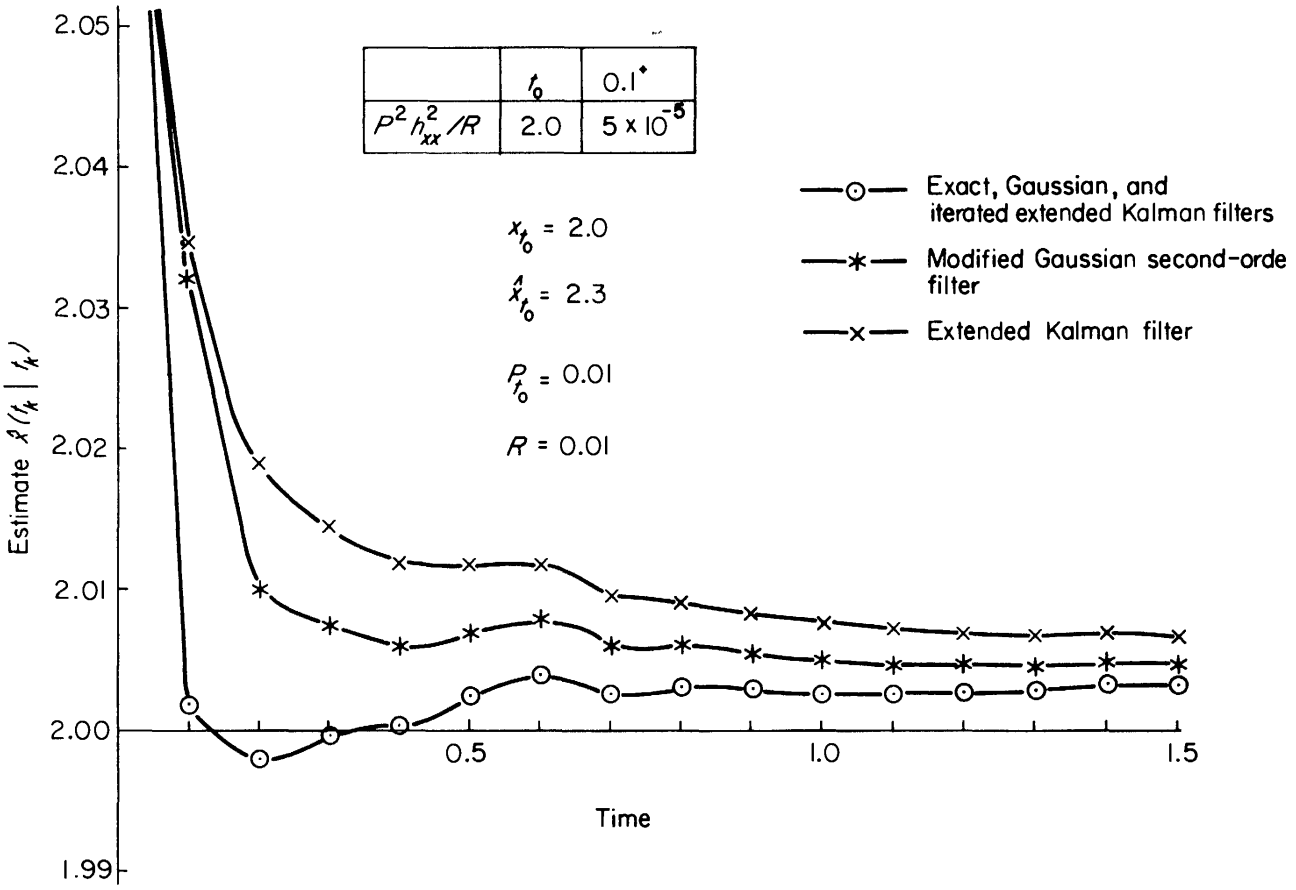


FIG. 9.2. Significance of measurement nonlinearity—II ($R = 0.01$).

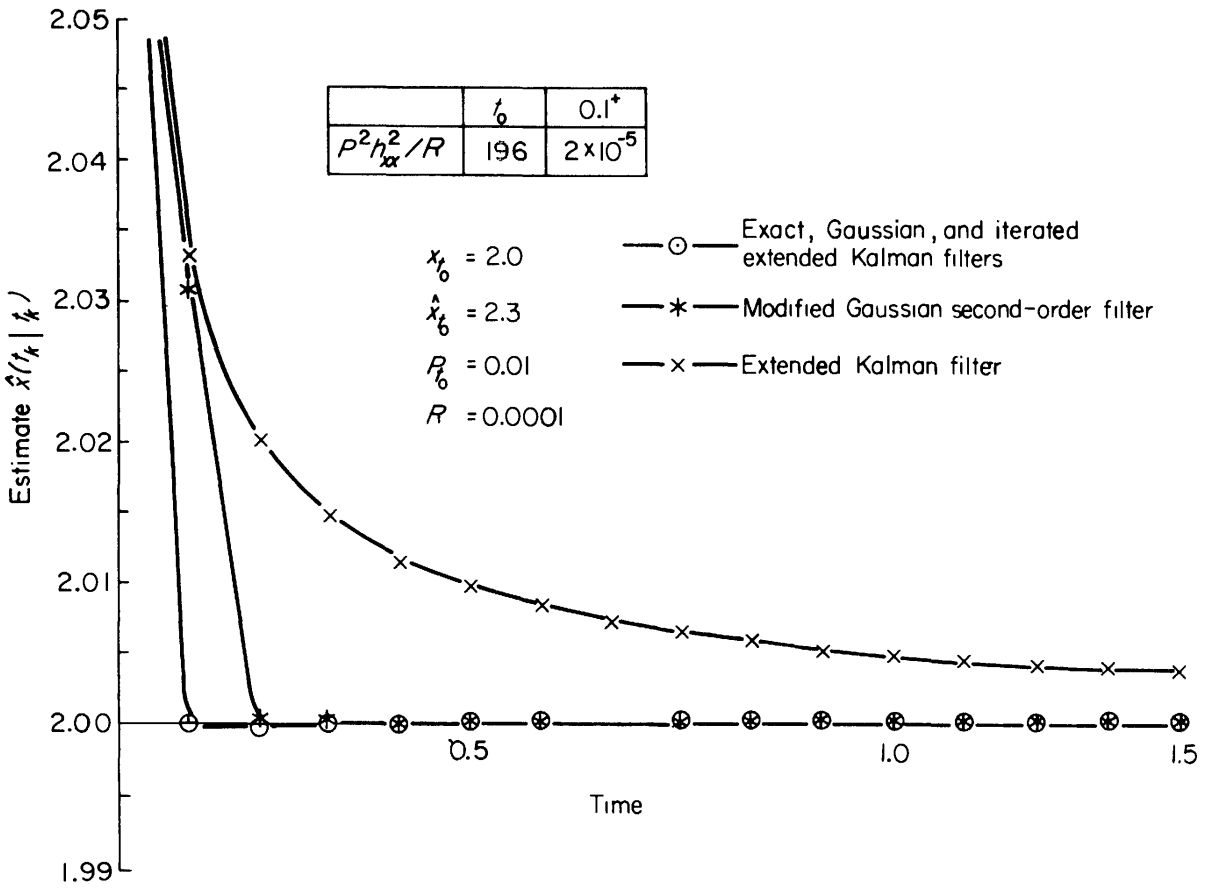


FIG. 9.3. Significance of measurement nonlinearity—III ($R = 0.0001$).

Fig. 9.1, where the measurement nonlinearity is relatively small ($P^2h_{xx}^2/R = 0.22$ initially), nonlinear effects are not very significant. Although the exact filter is initially best, subsequent performance of the various filters does not differ substantially. The extended Kalman filter would be chosen here as "best" on the basis of simplicity. As we go to larger measurement nonlinearities in Figs. 9.2 and 9.3, the picture changes significantly. The performance of the modified Gaussian second-order filter approaches that of the exact filter, whereas the extended Kalman filter estimate is biased. We note, however, that in all simulations the extended Kalman filter eventually converges to the true state. The larger the nonlinearity, the slower is the convergence of the extended Kalman filter. Simulations indicate that the modified Gaussian second-order filter adequately accounts for the biasing effect of measurement nonlinearities, although some bias still remains in cases where the measurement nonlinearity is marginally significant (see Fig. 9.2).

As we observed in Section 6, measurement nonlinearities are affected by h_{xx} and P_{t_0}. These effects are shown in Figs. 9.4 and 9.5. In Fig. 9.4, measurement nonlinearity is decreased relative to that in Fig. 9.3 by decreasing h_{xx}. In Fig. 9.5, the measurement nonlinearity is increased

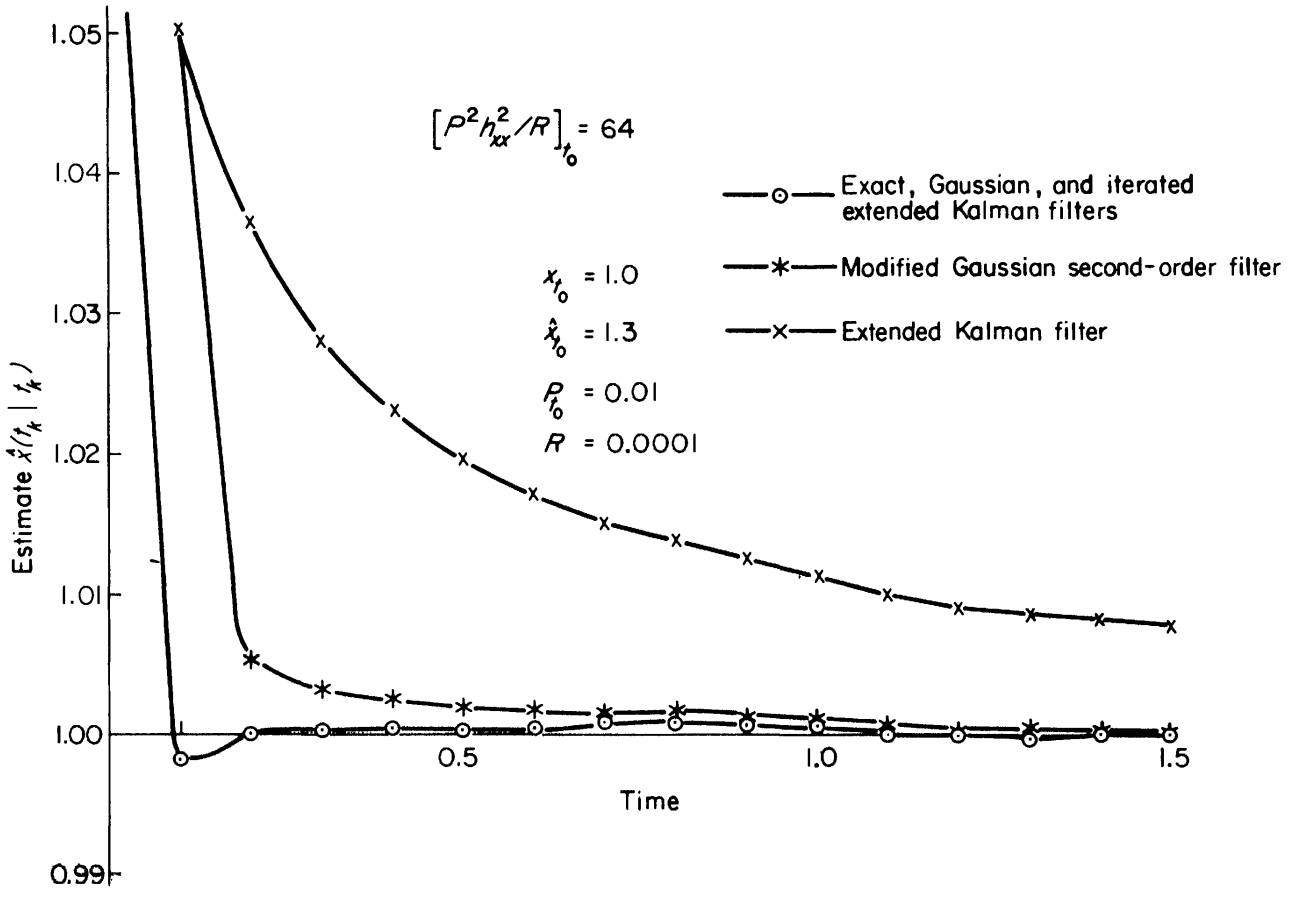


FIG. 9.4. Effect of h_{xx}

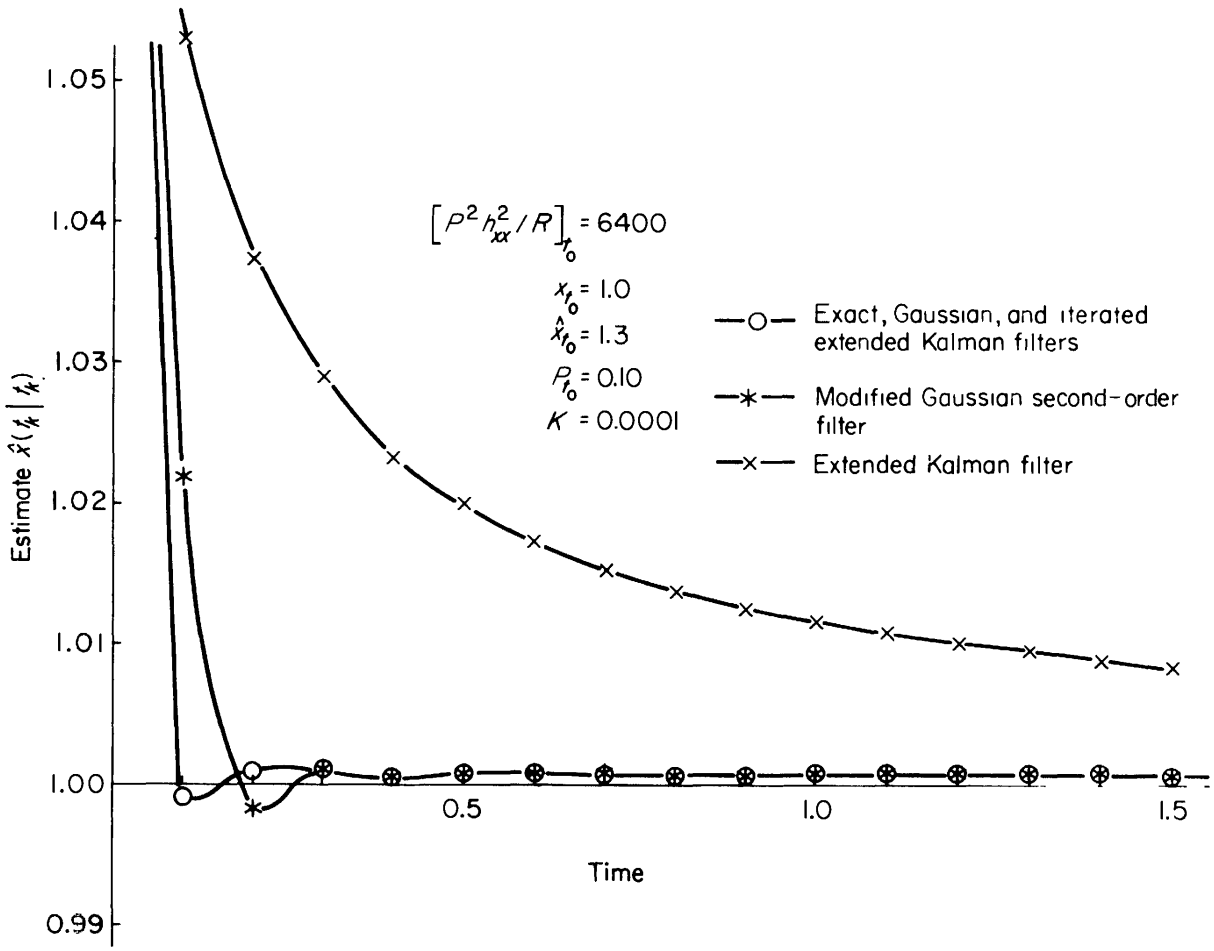


FIG. 9.5. Effect of P_{t_0}.

TABLE 9.1

CONDITIONAL VARIANCES ASSOCIATED WITH FIG. 9.4

	$P(t_k \mid t_k) \times 10^6$				
Time	Exact	Gaussian	IE–Kalman[a]	MGSO[b]	E–Kalman[c]
0.1	4.019	4.019	4.020	81.851	1.700
0.2	2.003	2.007	2.004	3.295	1.132
0.4	0.998	1.002	1.001	1.234	0.696
1.0	0.399	0.400	0.400	0.431	0.331
2.0	0.200	0.200	0.200	0.207	0.179

[a] Iterated extended Kalman filter.
[b] Modified Gaussian second-order filter.
[c] Extended Kalman filter.

relative to that in Fig. 9.4 by increasing P_{t_0}. The behavior of the nonlinear filters is as predicted, and as just described.

The conditional variances associated with the various filters, for the run in Fig. 9.4, are given in Table 9.1. Here we see again that the exact, Gaussian, and iterated extended Kalman filters are in excellent agree-

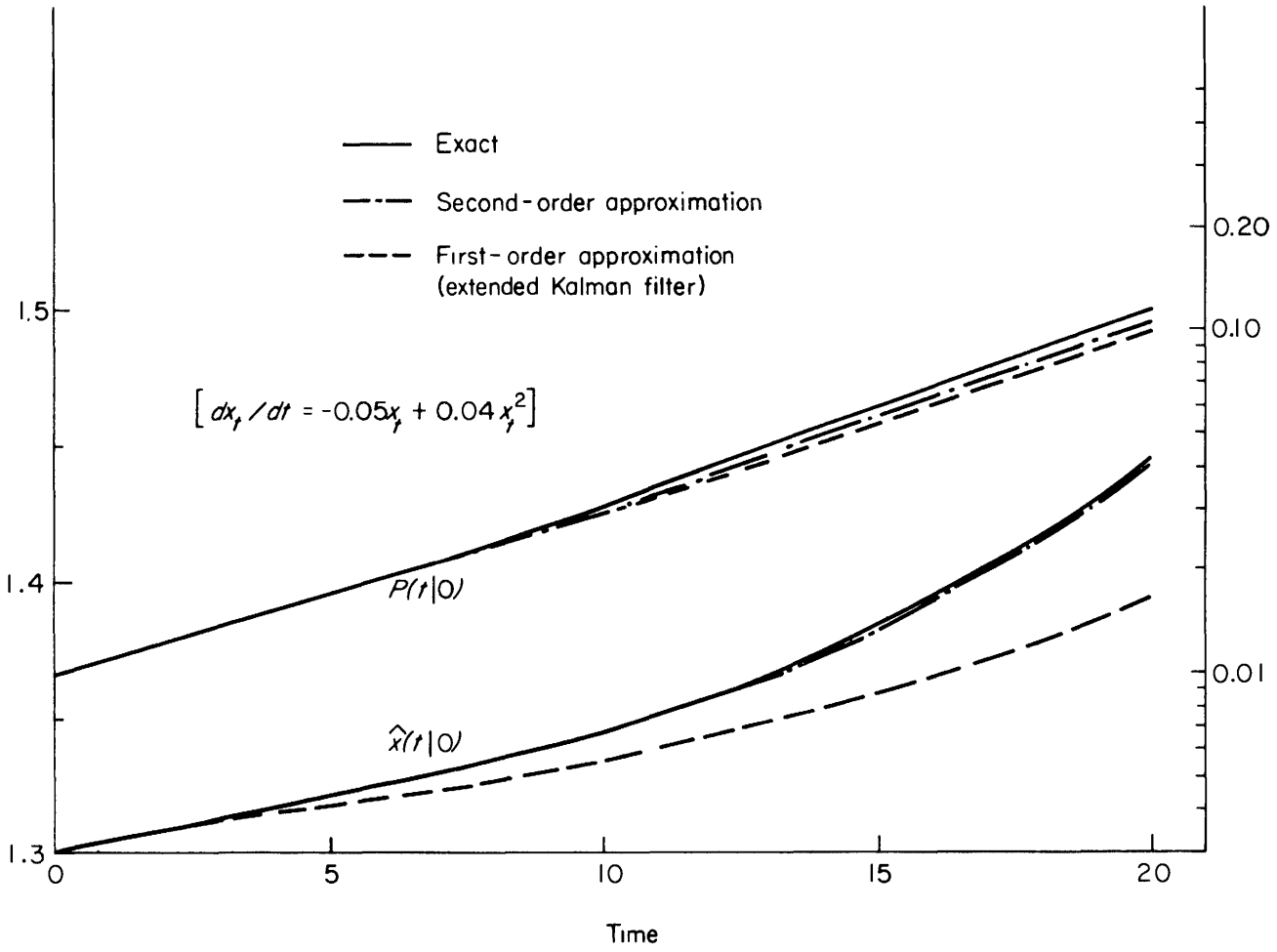


FIG. 9.6. System nonlinearities in prediction.

ment. The modified Gaussian second-order filter overestimates the variance, which is generally desirable. The extended Kalman filter usually underestimates the variance, which has a detrimental effect. The extended Kalman estimate is biased, but optimistic. This filter thinks it is performing very well, whereas, in reality, it is not.

When dynamics are introduced into the simulations we have described, the results remain essentially unchanged. The effect of system nonlinearities is not felt in the short time intervals between observations. The effect of measurements dominates.

The effect of system nonlinearities in (long-time) prediction is shown in Fig. 9.6. The effect on the conditional variance is not very significant, whereas the effect on the estimate itself is. First-order theory, which

neglects second-order nonlinearities, produces a significantly biased prediction. The second-order approximation accounts for most of this bias.

It should be noted that in our simulations the measurement nonlinearity decreased sharply after the first observation was processed. This is, of course, due to the sharp reduction in P. As a consequence, it is important to account for measurement nonlinearities early in the estimation. Subsequently, nonlinearities become insignificant, and the extended Kalman filter can then be used effectively. This can be seen in the simulation of the iterated extended Kalman filter. Generally, four iterations are required in processing the first observation. Subsequently, one iteration is usually sufficient. One iteration of the iterated extended Kalman filter is equivalent to the extended Kalman filter itself. This conclusion should not be generalized however. Nonlinearities can occur later in the estimation, as in the reentry filtering problem described in Section 10.

Simulations in our simple nonlinear example indicate that the iterated extended Kalman filter is extremely effective in accounting for measurement nonlinearities. The modified Gaussian second-order filter is also very effective, although it does not remove the biasing effect of nonlinearities completely. The first filter involves iteration, whereas the second one requires the computation of second partial derivatives of the measurement function. As a consequence, it is difficult to choose between the two on the basis of this example. We saw (Fig. 9.6) that the second-order approximation is very effective in removing the biasing effect of system nonlinearities. These nonlinear filters give significantly better performance than the extended Kalman filter.

9. APPLICATIONS IN ORBIT DETERMINATION

Extensive orbit determination experience at Goddard (see Wolf [30]) appears to indicate that the extended Kalman filter is adequate in this application. Second-order nonlinear effects seem to be negligible. This conclusion appears correct in view of this author's experiments with the modified Gaussian second-order filter in simple orbit determination problems.

We have in mind here orbit determination situations in which many data are available. Nonlinear effects in long-time prediction may still be significant. In that case, however, the effect of dynamic model errors is very important (see Chapter 8, particularly Section 8.8). The problem of model accuracy must clearly come first. It makes little sense to take

into account second derivatives of a function, when the function itself is not accurately known.

10. APPLICATIONS IN REENTRY

The first application of approximate nonlinear filters in a problem of practical significance containing significant nonlinearities was made by Athans, *et al.* [1], and Wishner *et al.* [29] in reentry filtering. The reader can refer to Section 8.7 and Appendix 8B for a description of reentry dynamics and filtering. We saw in Section 8.7 that nonlinearities are significant in reentry. They are particularly severe during high decelerations which occur midway in the reentry.

These authors simulated the extended Kalman filter, the iterated linear filter-smoother (Theorem 8.3), and the modified Gaussian second-order filter. Their problem is too complex to permit the computation of the exact conditional mean and covariance matrix, so that filter performance must be assessed on the basis of average properties. For this purpose, the average bias

$$b_j(t_k) = N^{-1} \sum_{i=1}^{N} [x_{t_k}^{(i)} - \hat{x}^{(i)}(t_k \mid t_k)]_j ,$$

and the *RMS* estimation error

$$RMS_j(t_k) = \left\{ N^{-1} \sum_{i=1}^{N} [(x_{t_k}^{(i)})_j - (\hat{x}^{(i)}(t_k \mid t_k))_j]^2 \right\}^{1/2}$$

for each filter are used. Here subscript j denotes the jth component of the vector, and superscript (i) denotes the ith simulation run.

These authors found that the iterated linear filter-smoother and the modified Gaussian second-order filter gave substantially better average performance than the extended Kalman filter. The improved performance was particularly striking in the high deceleration region and in the remainder of the reentry trajectory. In general, system nonlinearities appear much more significant than measurement nonlinearities in their examples. The modified Gaussian second-order filter successfully removes the biasing effect of nonlinearities and has the smallest average bias of the filters tested. On the other hand, the iterated linear filter-smoother shows somewhat better *RMS* error performance than the modified Gaussian second-order filter. This suggests the possibility of using the second-order filter (propagation of the conditional mean between observations) in the iterator.

In their simulations, the authors found that the modified Gaussian second-order filter consumed about 20% more computing time than the extended Kalman filter. Two iterations of the iterated filter of course consume twice as much time as the extended Kalman filter. Most of the performance improvement is fortunately achieved with two iterations.

11. SUMMARY

The extended Kalman filter has been successfully applied to numerous nonlinear filtering problems. If nonlinearities are significant, however, filter performance can be substantially improved by local iteration or inclusion of second-order effects. The modified Gaussian second-order filter has been applied in several problems with considerable success. The improvement in performance realized with the modified Gaussian second-order filter can be particularly significant in real-time applications where decisions must be made on the basis of the estimate and estimation error statistics. Such is the case in guidance and control applications.

It must be emphasized, however, that, in at least one highly nonlinear problem (see Kushner [19]), the modified Gaussian second-order filter diverged! The problems in which the modified Gaussian second-order filter performed well are problems in which the extended Kalman filter performed also, although perhaps not as well.

It is of course possible to develop higher-order approximations and to consider higher-order moments and even skewness (third moments). The practical usefulness of higher-order approximations is, however, questionable, especially in problems of high dimension (>3, say). Furthermore, it is questionable whether higher-order approximations would improve performance in cases where the extended Kalman filter does not work at all (diverges). The crucial defect in our approximations is that we replace global properties of a function (its average) by local properties (its derivatives). If the first-order approximation is completely inadequate, then finite higher-order approximation will tend to be useless as well.

In the final analysis, our approximations are *ad hoc* and must be tested by simulation. Thus, much more simulation experience with approximate nonlinear filters is highly desirable. Simulations could give considerable insight into the stability properties of these filters, as well as determine their statistical performance. Such simulations might also suggest different types of approximations.

Along theoretical lines, research is needed in techniques of parametrizing probability density functions.

Let us conclude by observing that, although approximate nonlinear filters have been found useful in some applications, their value in a particular problem cannot be assessed *a priori*, but must be determined by simulations.

APPENDIX 9A

APPROXIMATE CONTINUOUS FILTERS: VECTOR CASE

We develop in this appendix the truncated and Gaussian second-order filters for continuous vector systems. The approximations on which these filters are based are given in Section 3. They involve approximations to the expectations appearing in the equations for the conditional mean and covariance matrix in Theorem 6.6.

For both filters, we have

$$\hat{f}_i(x, t) = f_i(\hat{x}, t) + \frac{1}{2} \sum_{j,k=1}^{n} P_{jk} \frac{\partial^2 f_i(\hat{x}, t)}{\partial x_j \, \partial x_k}, \tag{9A.1}$$

$$\widehat{x_i h_j} - \hat{x}_i \hat{h}_j = \sum_{k=1}^{n} P_{ik} \frac{\partial h_j(\hat{x}, t)}{\partial x_k}, \tag{9A.2}$$

where we have used the symmetry of the covariance matrix P. Similar expressions hold for $\hat{h}_i$ and $(\widehat{x_i f_j} - \hat{x}_i \hat{f}_j)$. For the *truncated* second-order filter, we get

$$\begin{aligned} &[\widehat{x_i x_j h_k} - \widehat{x_i x_j} \hat{h}_k - \hat{x}_i \widehat{x_j h_k} - \hat{x}_j \widehat{x_i h_k} + 2\hat{x}_i \hat{x}_j \hat{h}_k] \\ &\quad = -\tfrac{1}{2} P_{ij} \sum_{q,r=1}^{n} \frac{\partial^2 h_k(\hat{x}, t)}{\partial x_q \, \partial x_r} P_{qr}, \end{aligned} \tag{9.A3}$$

whereas for the *Gaussian* second-order filter this quantity (in brackets) is

$$[\cdot] = \sum_{q,r=1}^{n} \frac{\partial^2 h_k(\hat{x}, t)}{\partial x_q \, \partial x_r} P_{iq} P_{jr}. \tag{9A.4}$$

Let us define the quantities

$$(P \, \partial^2 f)_i \triangleq \sum_{j,k=1}^{n} P_{jk} \frac{\partial^2 f_i(\hat{x}, t)}{\partial x_j \, \partial x_k}, \tag{9A.5}$$

$$(P^{\otimes 2} \, \partial^2 h)_{ijk} \triangleq \sum_{q,r=1}^{n} \frac{\partial^2 h_k(\hat{x}, t)}{\partial x_q \, \partial x_r} P_{iq} P_{jr}, \tag{9A.6}$$

and the matrices

$$\begin{aligned} F &\triangleq \left[\frac{\partial f_i(\hat{x}, t)}{\partial x_j}\right] \qquad (n \times n), \\ M &\triangleq \left[\frac{\partial h_i(\hat{x}, t)}{\partial x_j}\right] \qquad (m \times n), \\ (P\,\partial^2 f) &\triangleq [(P\,\partial^2 f)_1, \ldots, (P\,\partial^2 f)_n]^{\mathrm{T}} \qquad (n \times 1), \\ (P\,\partial^2 h) &\triangleq [(P\,\partial^2 h)_1, \ldots, (P\,\partial^2 h)_m]^{\mathrm{T}} \qquad (m \times 1). \end{aligned} \tag{9A.7}$$

Furthermore, if ξ is an m-vector, we define the operation

$$(P^2\,\partial^2 h) : \xi \triangleq \left[\sum_{k=1}^{m} (P^2\,\partial^2 h)_{ijk}\,\xi_k\right], \tag{9A.8}$$

which is an $n \times n$ symmetric matrix.

Then, from Theorem 6.6, we have the *truncated* second-order filter

$$\begin{aligned} d\hat{x}_t = {} & [f(\hat{x}_t, t) + \tfrac{1}{2}(P_t\,\partial^2 f)]\,dt \\ & + P_t M^{\mathrm{T}} R^{-1}[dz_t - (h(\hat{x}_t, t) + \tfrac{1}{2}(P_t\,\partial^2 h))\,dt], \end{aligned} \tag{9A.9}$$

$$\begin{aligned} dP_t = {} & [FP_t + P_t F^{\mathrm{T}} + \widehat{GQG}^{\mathrm{T}} - P_t M^{\mathrm{T}} R^{-1} M P_t]\,dt \\ & - \tfrac{1}{2} P_t\{(P_t\,\partial^2 h)^{\mathrm{T}}\, R^{-1}[dz_t - (h(\hat{x}_t, t) + \tfrac{1}{2}(P_t\,\partial^2 h))\,dt]\}. \end{aligned} \tag{9A.10}$$

Note that the quantity in braces in (9A.10) is a scalar. The *Gaussian* second-order filter has Eq. (9A.9) for the conditional mean, but has the variance equation

$$\begin{aligned} dP_t = {} & [FP_t + P_t F^{\mathrm{T}} + \widehat{GQG}^{\mathrm{T}} - P_t M^{\mathrm{T}} R^{-1} M P_t]\,dt \\ & + (P_t^2\,\partial^2 h) : \{R^{-1}[dz_t - (h(\hat{x}_t, t) + \tfrac{1}{2}(P_t\,\partial^2 h))\,dt]\}. \end{aligned} \tag{9A.11}$$

We leave it for the reader to work out $[GQG^{\mathrm{T}}]\hat{}$, which is different for the two filters.

APPENDIX 9B

APPROXIMATE CONTINUOUS-DISCRETE FILTERS: VECTOR CASE

In this appendix, we derive the truncated, Gaussian, and modified second-order filters for continuous-discrete vector systems. The approximations involved in these filters are described in Section 3.

The approximate equations of evolution for the conditional mean and covariance matrix *between observations* are the vector analogs of (9.17), obtained from (9A.9) and (9A.10) by setting $R^{-1} \equiv 0$:

$$\begin{aligned} d\hat{x}_t/dt &= f(\hat{x}_t, t) + \tfrac{1}{2}(P_t\, \partial^2 f), \\ dP_t/dt &= FP_t + P_t F^{\mathrm{T}} + \widehat{GQG^{\mathrm{T}}}. \end{aligned} \tag{9B.1}$$

The n-vector $P_t\, \partial^2 f$ and the $n \times n$ matrix F of partial derivatives are defined in Appendix 9A. The evaluation of $[GQG^{\mathrm{T}}]\hat{}$ for the truncated and Gaussian approximations is left to the reader [see (9.8) and (9.9) for the scalar case].

At an observation, we assume the form

$$\begin{aligned} \hat{x}(t_k \mid t_k) &= a + B(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}}), \\ [P(t_k \mid t_k)]_{ij} &= C_{ij} + \sum_{l=1}^{m} D_{ijl}(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})_l . \end{aligned} \tag{9B.2}$$

Here a is an n-vector, B is $n \times m$, and C is $n \times n$ with elements C_{ij}. Equations (9.23)–(9.25), which result from the application of (9.21), hold in the nonlinear case, except that now

$$y_{t_k} - \hat{y}_{t_k}^{t_{k-1}} = h - \hat{h} + v_k . \tag{9B.3}$$

Using the notation described below Eq. (9.34), Eqs. (9.23)–(9.25) give

$$a = \hat{x}(t_k \mid t_{k-1}), \tag{9B.4}$$

$$B = (\widehat{xh^{\mathrm{T}}} - \hat{x}\hat{h}^{\mathrm{T}})[[(h - \hat{h})(h - \hat{h})^{\mathrm{T}}]\hat{} + R]^{-1}, \tag{9B.5}$$

$$C = P(t_k \mid t_{k-1}) - (\widehat{xh^{\mathrm{T}}} - \hat{x}\hat{h}^{\mathrm{T}})[[(h - \hat{h})(h - \hat{h})^{\mathrm{T}}]\hat{} + R]^{-1} (\widehat{hx^{\mathrm{T}}} - \hat{h}\hat{x}^{\mathrm{T}}). \tag{9B.6}$$

The coefficients D_{ijl} are evaluated by setting

$$s = [x_{t_k} - \hat{x}(t_k \mid t_k)]_i\, [x_{t_k} - \hat{x}(t_k \mid t_k)]_j ,$$

$$u = (y_{t_k} - \hat{y}_{t_k}^{t_{k-1}})_l$$

in (9.21), whereby we obtain

$$\begin{aligned} &[(x - \hat{x})_i\, (x - \hat{x})_j\, (h - \hat{h})_l]\hat{} - [(x - \hat{x})_i\, (B(h - \hat{h}))_j\, (h - \hat{h})_l]\hat{} \\ &\quad - [(B(h - \hat{h}))_i\, (x - \hat{x})_j\, (h - \hat{h})_l]\hat{} + [(B(h - \hat{h}))_i\, (B(h - \hat{h}))_j\, (h - \hat{h})_l]\hat{} \\ &\quad = \sum_{q=1}^{m} D_{ijq}[[(h - \hat{h})_q\, (h - \hat{h})_l]\hat{} + R_{ql}], \end{aligned}$$

which, after some manipulation, reduces to

$$D_{ijl} = \sum_{q=1}^{m} \Big\{ [(x - \hat{x})_i (x - \hat{x})_j h_q]\hat{} - \sum_{r=1}^{m} B_{jr}[(h - \hat{h})_r (x - \hat{x})_i h_q]\hat{}$$

$$- \sum_{r=1}^{m} B_{ir}[(h - \hat{h})_r (x - \hat{x})_j h_q]\hat{}$$

$$+ \sum_{r,p=1}^{m} B_{ir}B_{jp}[[(h - \hat{h})_r (h - \hat{h})_p h_q]\hat{} + R_{rp}\hat{h}_q] - C_{ij}\hat{h}_q \Big\}$$

$$\times \{[(h - \hat{h})(h - \hat{h})^{\mathrm{T}}]\hat{} + R\}_{ql}^{-1}. \tag{9B.7}$$

Since the coefficients D_{ijl} are extremely cumbersome, although straightforward, to evaluate, and because of the numerical results of Section 8, we shall not evaluate them here. The interested reader may evaluate them himself, or refer to Jazwinski [16]. We shall develop here the modified second-order filters of the form

$$\begin{aligned} \hat{x}(t_k \mid t_k) &= a + B(y_{t_k} - \hat{y}_{t_k}^{t_{k-1}}), \\ P(t_k \mid t_k) &= C. \end{aligned} \tag{9B.8}$$

For both the truncated and Gaussian approximations,

$$\widehat{xh}^{\mathrm{T}} - \hat{x}\hat{h}^{\mathrm{T}} = P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(t_k), \tag{9B.9}$$

where the matrix of partial derivatives M is defined in (9A.7). For the *truncated* approximation,

$$[(h - \hat{h})(h - \hat{h})^{\mathrm{T}}]\hat{} = M(t_k)\, P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(t_k) - \tfrac{1}{4}(P\,\partial^2 h)(P\,\partial^2 h)^{\mathrm{T}}, \tag{9B.10}$$

where the vector $P\,\partial^2 h$ is defined in (9A.7); whereas for the *Gaussian* approximation,

$$[(h - \hat{h})(h - \hat{h})^{\mathrm{T}}]\hat{} = M(t_k)\, P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(t_k) + \tfrac{1}{2}(\partial^2 h P^2\, \partial^2 h), \tag{9B.11}$$

where $\partial^2 h P^2\, \partial^2 h$ is the symmetric $m \times m$ matrix with elements (i, j)

$$\sum_{k,l,p,q=1}^{n} \frac{\partial^2 h_i}{\partial x_k\, \partial x_l} P_{lp} P_{kq} \frac{\partial^2 h_j}{\partial x_p\, \partial x_q} \qquad [P = P(t_k \mid t_{k-1})]. \tag{9B.12}$$

Now, defining the $m \times m$ matrices,

$$Y^t \triangleq M(t_k)\, P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(t_k) + R(k) - \tfrac{1}{4}(P\,\partial^2 h)(P\,\partial^2 h)^{\mathrm{T}}, \tag{9B.13}$$

$$Y^G \triangleq M(t_k)\, P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(t_k) + R(k) + \tfrac{1}{2}(\partial^2 h P^2\, \partial^2 h), \tag{9B.14}$$

the *modified truncated second-order filter* becomes (at an observation)

$$\hat{x}(t_k \mid t_k) = \hat{x}(t_k \mid t_{k-1}) + P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(t_k)[Y^t]^{-1} \times [y_{t_k} - h(\hat{x}(t_k \mid t_{k-1}), t_k) - \tfrac{1}{2}(P(t_k \mid t_{k-1})\, \partial^2 h)], \tag{9B.15}$$

$$P(t_k \mid t_k) = P(t_k \mid t_{k-1}) - P(t_k \mid (t_{k-1})\, M^{\mathrm{T}}(t_k)[Y^t]^{-1}\, M(t_k)\, P(t_k \mid t_{k-1}).$$

The *modified Gaussian second-order filter* is given by

$$\hat{x}(t_k \mid t_k) = \hat{x}(t_k \mid t_{k-1}) + P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(t_k)[Y^G]^{-1} \times [y_{t_k} - h(\hat{x}(t_k \mid t_{k-1}), t_k) - \tfrac{1}{2}(P(t_k \mid t_{k-1})\, \partial^2 h)], \tag{9B.16}$$

$$P(t_k \mid t_k) = P(t_k \mid t_{k-1}) - P(t_k \mid t_{k-1})\, M^{\mathrm{T}}(t_k)[Y^G]^{-1}\, M(t_k)\, P(t_k \mid t_{k-1}).$$

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